

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		A and C and E identified as correct (2) 2 correct gains one mark -1 for each additional tick > 3 ticks	2			2		
	(b)	(i)	$x = (\cos 25^\circ \times 12) - 5.44$ seen or $x = 5.44$ or $6 \cos 25^\circ$ or use of $\sin 25$ and Pythagoras or argument based on C of G of extendable arm in middle therefore $x = 2.00 + 3.44$		1		1	1	
		(ii)	C.M. = $2.2 \times 10^4 \times 2.00$ (or 4.4×10^4 seen) (1) A.C.M. = $(W \times 8.88 \text{ ecf}) + (3.0 \times 10^3 \times 3.44)$ (1) Correct algebra and W shown = 3793 [N] (1) 3767 [N] if approximation of 5.5 used 3810 [N] if approximation of 5.4 used		3		3	3	
		(iii)	3800 ecf – 1700 = 2100 [N] (1) Conversion of mass-force i.e. $m = \frac{F}{g}$ (1) = 214 [kg] (1) Conclusion reasonable and consistent with their figures (1) Alternative: 2 people mass approx 200 kg (1) Conversion of mass-force i.e. $F = mg$ (1) Adding both forces – giving around 3700 [N] depending on mass (1) Conclusion reasonable and consistent with their figures ecf on comparison with (b)(ii) (1)			4	4	2	
			Question 2 total	2	4	4	10	6	0